GRPO-NS: Group Relative Policy Optimization with Negative Sampling for Diverse, Targeted de-novo Drug Generation

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Abstract

We propose GRPO-NS, a reinforcement-learning framework that adapts Group Relative Policy Optimization (GRPO) to sequence-based molecular generators and augments it with negative sampling (NS) for improved reward shaping, robustness, and diversity. GRPO, originally developed to efficiently fine-tune large language models by comparing groups of candidate outputs instead of relying on a learned critic, offers sample-efficient policy updates well suited to expensive evaluation functions common in drug discovery.

1 Introduction

De-novo molecular generation aims to design novel compounds with optimal pharmacological properties, a key challenge in computational drug discovery. While RL methods like PPO [1, 2, 5]and actor-critic approaches optimize molecular rewards, progress is limited by two issues: (1) sparse, expensive rewards from costly evaluations (e.g., docking, quantum simulations) and (2) mode collapse, where the model produces repetitive, low-diversity molecules.

Recent advances like GFlowNets [3, 4, 7] address molecular diversity by sampling molecules proportional to their rewards. Group Relative Policy Optimization (GRPO), developed for large language models, removes the critic by using group-relative ranking, improving stability and sample efficiency in noisy reward environments—yet it remains unexplored in molecular design.

Negative sampling, which introduces low-quality examples during training, helps models identify undesirable regions of chemical space. Combining these ideas, GRPO-NS aims to balance exploration and exploitation: GRPO ensures stable learning under sparse rewards, while negative sampling improves discrimination and reduces reward hacking.[2, 5].

In this work, we introduce GRPO-NS, a hybrid approach designed to maximize the diversity and quality of generated drug-like molecules. We evaluate its performance on multiple benchmark tasks (e.g., QED, SAScore, predicted binding affinity) and compare it against PPO and GFlowNet [3, 4, 7]baselines. Through extensive experiments and ablation studies, we aim to demonstrate that GRPO-NS achieves a superior tradeoff between optimization efficiency, chemical diversity, and robustness, providing a promising direction for RL-driven drug design.[1, 2]

2 Problem Description

De-novo molecular generation aims to propose novel small molecules with desirable properties (binding affinity, ADMET profile, synthesizability). Two recurring practical problems slow progress:

Sparse and expensive rewards — high-quality evaluations (docking, wet-lab assays) are slow and noisy; surrogate models introduce bias.

Mode collapse / low diversity — policy gradients optimizing a narrow reward often produce many similar molecules, limiting hit discovery.[6]

The system implements a simplified RLUF approach using direct verifiable rewards (+1 for correct answers, -0.1 for incorrect) to eliminate the complexity and cost of traditional reward models while maintaining learning capabilities. Students interact with the AI tutor, receive personalized explanations, take follow-up quizzes, and their performance automatically trains the system to provide better future responses. This creates a continuous improvement loop where the AI becomes increasingly effective at teaching each individual student based on their demonstrated learning patterns and comprehension levels.

3 *Current Limitations*

- Static Teaching Methods: Existing AI tutors cannot adapt their explanations or questioning strategies based on whether students actually learn from their responses, resulting in repeated ineffective teaching approaches.
- Lack of Deep Personalization: Current systems fail to integrate comprehensive student profiles (learning style, demographics, academic background) into real-time responses, providing generic rather than individually tailored educational experiences.
- No Learning from Outcomes: Traditional AI tutors cannot optimize their teaching effectiveness based on student comprehension and retention data, missing opportunities to improve educational outcomes through iterative refinement.
- High Implementation Costs: Complex reward model systems require expensive computational resources ($50,000+) and extensive development time (8-12 weeks), making advanced AI tutoring inaccessible to many educational institutions.
- Generic Content Delivery: Most AI tutors draw from general knowledge bases rather than curriculum-specific, age-appropriate educational content, leading to responses that may be technically correct but ineffective for specific learning objectives and developmental stages.

4 Technical Proposal

4.1 Technical Approach

4.1.1 *System Architecture*

Our adaptive AI tutor implements a streamlined three-component architecture: (1) Merlyn Mind's 12B education-specific language model as the foundation [6] (2) Dynamic prompt generation engine that injects comprehensive user profiles into real-time interactions (3) Direct RLUF training pipeline using verifiable quiz rewards (+1 correct, -0.1 incorrect) without complex reward models [3, 9]. The system leverages Knowledge Graph-enhanced Retrieval-Augmented Generation (KG-RAG) to ground responses in structured educational content, eliminating hallucination issues common in general-purpose models [8]. User profiles containing demographics, learning style, academic history, and performance patterns are dynamically integrated into personalized prompts, creating individualized educational experiences that adapt continuously based on quiz outcomes [1].

4.1.2 *Reinforcement Learning Implementation*

We implement Proximal Policy Optimization (PPO) with direct verifiable rewards, eliminating the need for separate reward model training [3, 7]. The system processes student interactions through a continuous feedback loop: tutor response → quiz administration → performance evaluation → reward calculation → model update [9].

4.1.3 *Personalization Framework*

The system builds comprehensive learner models incorporating: academic background, learning preferences, demographic factors, performance history, and real-time engagement metrics [1, 2]. These profiles are injected directly into model prompts, enabling context-aware responses that adapt teaching style, difficulty level, and content presentation to individual student needs without requiring separate personalization models [1, 5].

4.2 Evaluation Methods

- Quiz Performance Improvement: Pre/post assessment comparison with target 25-35% improvement in accuracy [1, 2].
- Knowledge Retention Testing: Follow-up assessments at 1-week and 1-month intervals to measure long-term learning [1, 2].
- Engagement Analytics: Session duration, return rate, and voluntary interaction frequency [1, 3].

5 Related Work

Reinforcement Learning from User Feedback (RLUF) represents a paradigm shift in AI alignment, moving from expert-annotated preferences to direct user signals [4]. Han et al. (2025) demonstrated that RLUF using binary feedback (like ”love” reactions) achieves 28% improvement in positive user engagement while maintaining safety standards [3].

Intelligent Tutoring Systems (ITS) research shows significant promise for personalized education [1, 2]. Létourneau et al. (2025) found that AI-driven ITS improve K-12 learning outcomes by 25-40% through adaptive learning pathways and real-time feedback mechanisms [2]. Contemporary studies emphasize that effective personalization requires comprehensive learner modeling incorporating cognitive states, learning styles, and performance patterns [1]. Liu et al. (2024) demonstrated that adaptive prompts in ITS, grounded in educational theories, enhance problem-solving efficiency and deepen understanding through dynamic adjustment to student performance [5].

Adaptive Learning Systems utilizing AI have shown substantial educational benefits [1]. Wang et al. (2024) established that incorporating detailed user profiles in educational AI systems leads to 35% improvement in learning efficiency [1]. Recent systematic reviews highlight that AI-enabled adaptive learning systems achieve superior outcomes through continuous assessment, real-time adjustments, and multi-objective optimization balancing engagement, comprehension, and retention [1, 2]. However, current literature identifies critical gaps in systems that can learn from educational outcomes to optimize teaching effectiveness over time, representing the key innovation opportunity addressed by our proposed RLUF-enhanced tutoring system

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